

Assignment 1 - Basic Cybersecurity Risk Assessment

To: BrightLayer Stores Management

From: Junior Cybersecurity Analyst

Subject: Cybersecurity Risk Assessment

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FIVE IMPORTANT ASSETS

1. **Employee email accounts** – Used for business communication, customer interactions, and potentially accessing other company services. Compromised accounts could expose sensitive information.
2. **Shared computers** – Used by employees to perform daily business activities. They may contain business files, saved credentials, and access to company systems.
3. **Company website** – Represents BrightLayer Stores publicly and may be used by customers to obtain information or interact with the business. Website compromise could damage the company's reputation.
4. **Cloud storage with invoices** – Contains important financial and business records. Loss, theft, or unauthorized modification of invoices could cause financial and operational problems.
5. **User passwords** – Protect employee accounts and company resources. If passwords are stolen or weak, attackers could gain unauthorized access.

2. THREE POTENTIAL THREATS

- **Phishing attackers** – Attackers may send fake emails to employees to steal login credentials or gain access to company accounts.
- **Malware/cyber attackers** – Attackers may use malicious software to compromise shared computers, steal information, or disrupt business operations.
- **Zero-day attackers** – An attacker could exploit a previously unknown security flaw in the company's website or software before a security patch is available.

3. THREE VULNERABILITIES

1. **Weak or reused passwords** – Employees may use passwords that are easy to guess or reuse across multiple accounts.
2. **Unpatched software and systems** – Shared computers or the company website may contain outdated software with security weaknesses. This could also expose the company to exploitation of a zero-day vulnerability.
3. **Poorly secured Wi-Fi router** – Weak router settings, outdated firmware, or inadequate network security could allow unauthorized users to access the network.

4. ASSOCIATED RISKS

1. **Phishing + weak or reused passwords:** An attacker could steal employee credentials and access email or cloud storage, potentially exposing sensitive business and invoice information.
2. **Malware + unpatched shared computers:** Malware could result in data theft, file loss, system downtime, or disruption to business operations.
3. **Zero-day attack + vulnerable or unpatched website or software:** An attacker could gain unauthorized access to the website or systems before a patch is available, potentially causing data loss or service disruption.
4. **Wi-Fi attack + poorly secured router:** An unauthorized user could gain access to the company's network and potentially intercept or access business information.

5. SECURITY RECOMMENDATIONS

1. Enable multi-factor authentication (MFA) for employee email, cloud storage, and other important accounts to reduce the impact of stolen passwords.
2. Implement strong, unique passwords and use a password manager to prevent password reuse.

3. Keep operating systems, applications, website software, and router firmware updated and establish a regular patching process.
4. Install endpoint protection on shared computers and restrict unnecessary software and user permissions.
5. Secure the Wi-Fi network using strong encryption, a strong administrator password, updated firmware, and a separate guest network.
6. Provide phishing awareness training so employees can recognize suspicious emails and avoid sharing credentials.
7. Limit access to cloud invoices based on employee roles and ensure important data is regularly backed up.

